Local projections: Comparison with Structural VARs

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Outline

1. LP vs. VAR
2. Identification
3. Bias-variance trade-off
4. Uncertainty assessments
5. Long horizons and persistent data

Impulse response estimators

- Both LPs and SVARs estimate structural impulse responses (dynamic causal effects):

$$\theta_h = E[y_{t+h} \mid \varepsilon_t = 1] - E[y_{t+h} \mid \varepsilon_t = 0], \quad h = 0, 1, 2, \ldots$$

  - Not a forecast. Shock ε_t may only explain small fraction of variation.

- Both methods are widely used, but they seem conceptually quite different.

  1. VAR: iterate on dynamic multivariate model. Sims (1980, 21.5k cites)
  2. LP: direct regression of future outcome y_{t+h} on current covariates. Jordà (2005, 4.5k cites)

- Which one works best?

LP or VAR? Preview of results

- LP & VAR are two *estimation* methods. Either method can be used to implement any given *identification* scheme (short-run/long-run restrictions, proxies, etc.).
- Must navigate a stark bias-variance trade-off:
  - LP: low bias, high variance.
  - VAR (few lags): potentially high bias, low variance. More lags $\Rightarrow$ closer to LP.
- For reliable confidence bands, choose (a) LP or (b) VAR with very many lags.

Local projection

- LP: linear regression, separately for each horizon $h = 0, 1, 2, \ldots$:

$$y_{t+h} = \mu_h + \theta_h^{\text{LP}} x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}.$$

  - This is a projection, not a generative model.

- **Shock:** LP estimates impulse response of y_{t+h} with respect to

$$\tilde{x}_t = x_t - \text{proj}(x_t \mid r_t, w_{t-1}, \ldots, w_{t-p}).$$

  Whether this is an interesting object depends on our assumptions, as we've discussed.

- **Projection:** LP uses autocorrelations in the data out to the horizon h of interest to compute θ_h^{LP}.

Vector autoregression

- VAR: estimate reduced-form multivariate dynamic model

$$w_t = c + A_1 w_{t-1} + A_2 w_{t-2} + \cdots + A_p w_{t-p} + u_t.$$

- Orthogonalize $u_t = H\varepsilon_t$. For now, assume H lower triangular (recursive/Cholesky id'n).
- Structural impulse responses $\Psi_h = \partial w_{t+h}/\partial \varepsilon_t'$ from iterative propagation:

$$\Psi_0 = H, \quad \Psi_1 = A_1\Psi_0, \quad \Psi_2 = A_1\Psi_1 + A_2\Psi_0, \quad \ldots \quad \Psi_h = \textstyle\sum_{\ell=1}^{\min\{p,h\}} A_\ell \Psi_{h-\ell},$$

$$\theta_h^{\text{VAR}} = \partial y_{t+h}/\partial \varepsilon_{x,t} = e_y' \Psi_h e_x.$$

- **Shock:** residual in projection of $u_{x,t}$ on $u_{r,t}$ (elements of u_t). Same as LP shock $\tilde{x}_t$!
- **Projection:** VAR matches first p autocovariances of the data, but extrapolates to longer horizons $h > p$.

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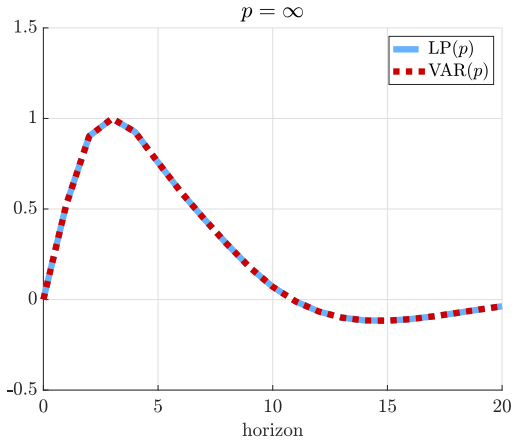
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Equivalence between LP and VAR

- LP & VAR project on the exact same shock.
- But the way they propagate that shock differs: LP directly uses autocorrelations in the data out to horizon h, while VAR extrapolates from the first p autocovariances.
- This suggests that if p is large, the two methods should be equivalent.
- Even if p is not large, the two methods should give similar results at horizons $h \leq p$.

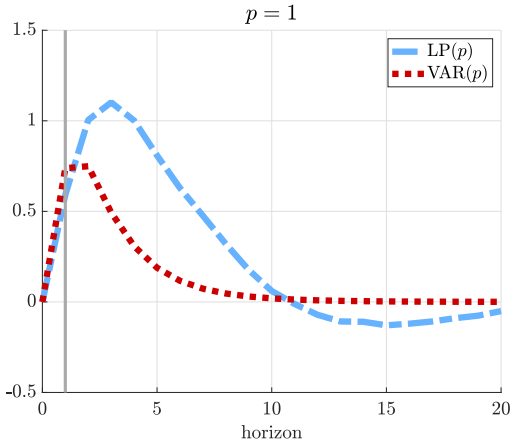
LP = VAR with very long lag length

$p = \infty$: same shock, same projection, so same impulse responses



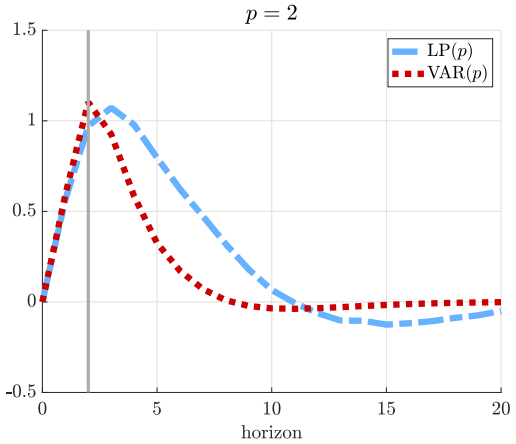
LP $\approx$ VAR up to horizon p

$p < \infty$: same shock so same responses at $h = 0$,
approx'ly same for $0 < h \leq p$, but then VAR extrapolates



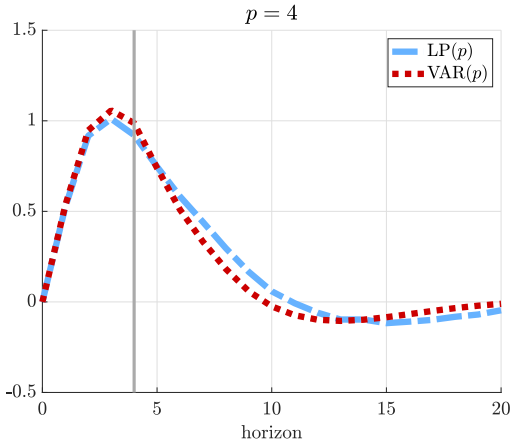
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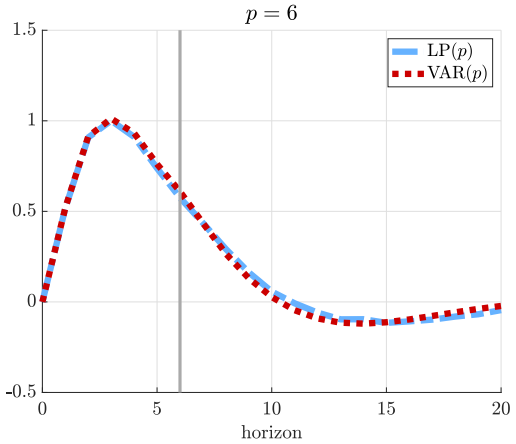
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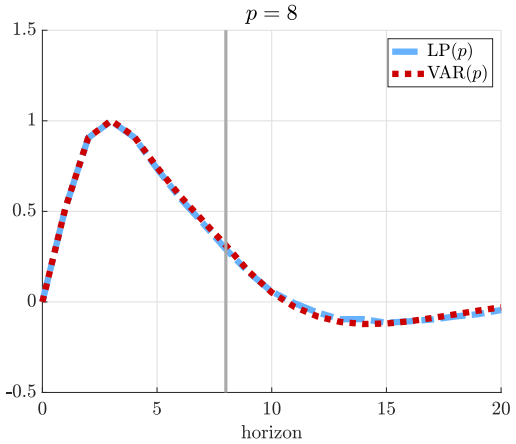
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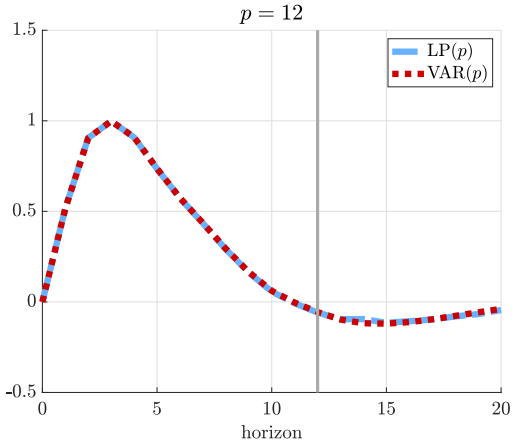
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LPs and VARs share the same estimand

- The previous discussion considered recursive identification (short-run timing restrictions).
- The equivalence between LP & VAR extends to more complicated identification schemes.
  - Proxy/IV, long-run restrictions, sign restrictions, ...
  - Intuition: "shock" is still just some (potentially complicated) f'n of autocovariances of the data. With many lags, both LP and VAR approximate these well in large samples.
- Take-away: LP vs. VAR debate unrelated to questions of identification.
  - Any identification scheme that can be implemented by an SVAR can also be implemented by an LP, and *vice versa*.
  - Only difference is how a finite data set is exploited to estimate the common estimand.

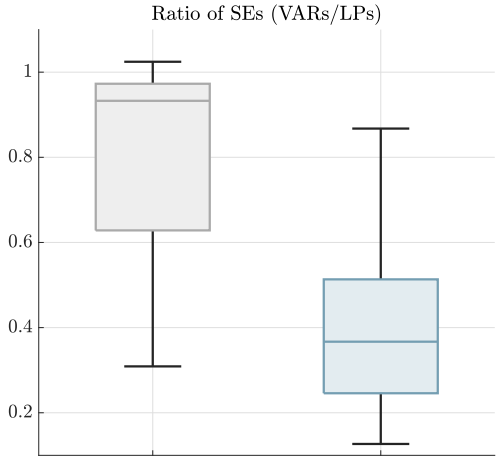
Proxy/IV identification

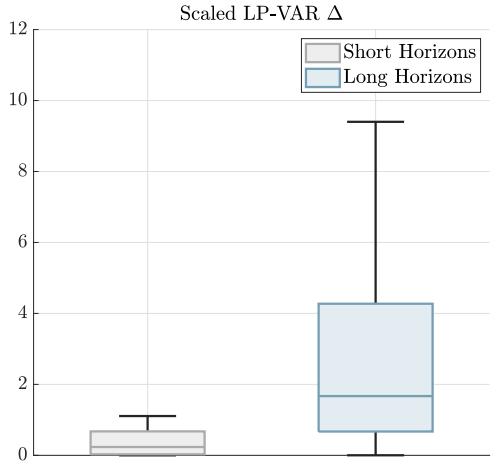
- For the case of identification via proxy/IV, the LP-IV estimator is equivalent (in large samples and with many lags) with an "internal instrument" SVAR estimator:
  - Include the proxy z_t directly in the list of variables in the SVAR, and order it *first*.
  - Compute Cholesky-orthogonalized impulse responses with respect to the *first shock*.
  - Scale the magnitude of the shock using the same unit effect normalization as for LP-IV.
- Differs from "external instrument" SVAR (Stock & Watson, '12; Mertens & Ravn, '13):
  - Estimate a VAR that *excludes* the proxy z_t.
  - Only use the proxy subsequently to identify the underlying structural shock.
- The latter procedure is less robust, as it requires the additional assumption that the shock is "invertible" (Stock & Watson, 2018; Plagborg-Møller & Wolf, 2021).

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VAR vs. LP in finite samples

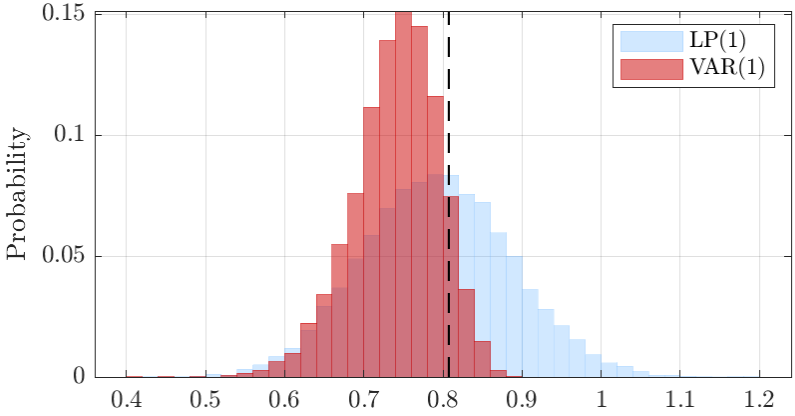




Replication of 4 empirical applications in Ramey (2016), total of 385 impulse responses

Illustrative simulation

$$y_t = \rho y_{t-1} + \varepsilon_t + \alpha \varepsilon_{t-1}, \quad \varepsilon_t \overset{i.i.d.}{\sim} N(0,1)$$



$h = 2,\ \rho = 0.85,\ \alpha = 0.1,\ T = 240$

Analytics of the bias-variance trade-off

- Consider a structural VAR model contaminated by small MA terms:

$$w_t = A_1 w_{t-1} + \cdots + A_{p_0} w_{t-p_0} + H\left(\varepsilon_t + \alpha_1 \varepsilon_{t-1} + \alpha_2 \varepsilon_{t-2} + \ldots\right).$$

  - Why? Low-order VARs are known to deliver good forecasts, but not literal truth.
  - MA terms can arise if we slightly under-specify the lag length, forget to control for a relevant variable, aggregate the data inappropriately, or use data with measurement error.
  - Technically, assume $\alpha_\ell \propto$ std. dev. of VAR estimator.

- In this environment, estimators should control for infinitely many lags. Infeasible.

Analytics of the bias-variance trade-off

- Suppose both LP & VAR use $p \geq p_0$ estimation lags.
- Then in large samples,

$$\hat{\theta}_h^{\text{VAR}} \dot{\sim} N\left(\theta_h + b_h(p), \tau^2_{h,\text{VAR}}(p)\right), \quad \hat{\theta}_h^{\text{LP}} \dot{\sim} N\left(\theta_h, \tau^2_{h,\text{LP}}\right).$$

  - Benefit and cost of extrapolation: VAR more efficient ($\tau^2_{h,\text{VAR}}(p) \leq \tau^2_{h,\text{LP}}$) but biased.
  - $h \leq p - p_0$: VAR bias $b_h(p) = 0$ and variance both coincide with LP.
- Both LP & VAR require controlling for the most important predictors/lags. But the result implies that LP is robust to omitting moderately important ones, while VAR is not.

How bad can the VAR bias be in theory?

- Theoretical bound on bias: letting $\mathcal{M}$ denote the fraction of the variance of the MA residual that's due to lagged shocks,

$$|b_h(p)| \leq \sqrt{T \times \mathcal{M} \times \left\{\tau^2_{h,\text{LP}} - \tau^2_{h,\text{VAR}}(p)\right\}},$$

  and there exist MA coefficients that attain the bound.

  - Example: if $T = 100$, $\mathcal{M} = 1\%$, $\tau_{h,\text{VAR}}(p)/\tau_{h,\text{LP}} = 0.5$, then bias can be $1.73 \times$ SE.

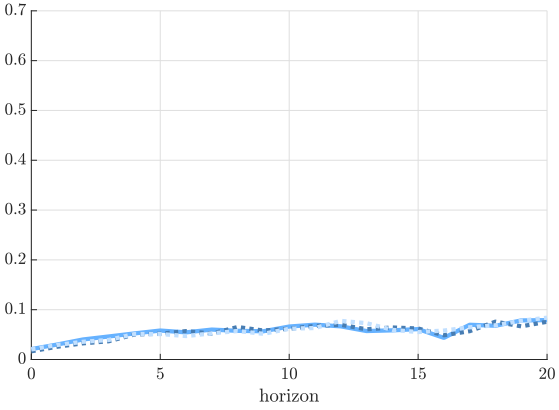
- No free lunch for VARs: if precision gain is large, then so is the potential bias.

  - VAR only robust if we use so many lags that VAR = LP.

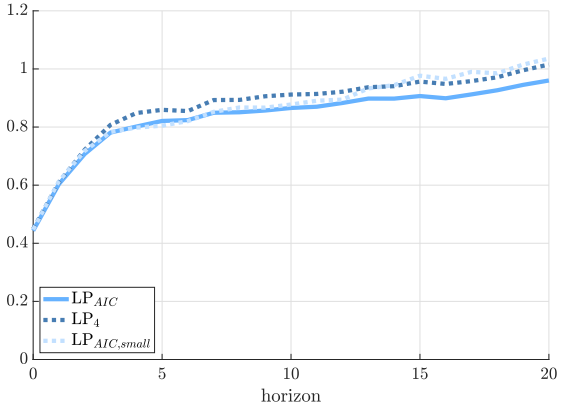
The bias-variance trade-off in practice

- Montiel Olea et al. (2025) conduct large-scale simulation study, extending Li et al. (2024).
- DGP: extension of Stock-Watson dynamic factor model fitted to 207 U.S. macro series.
  - Both stationary and non-stationary versions.
- Construct 100s of specifications:
  - Randomly draw subsets of 5 salient macro series from the DFM. Outcome y chosen at random from this list.
  - Additionally, econometrician observes a monetary/fiscal shock.
- Simulate data with $T = 240$, then estimate LPs, VARs, and several variants.

Simulation evidence: bias and standard deviation



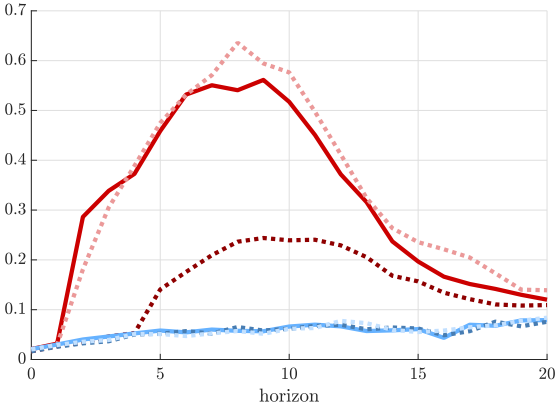
Bias



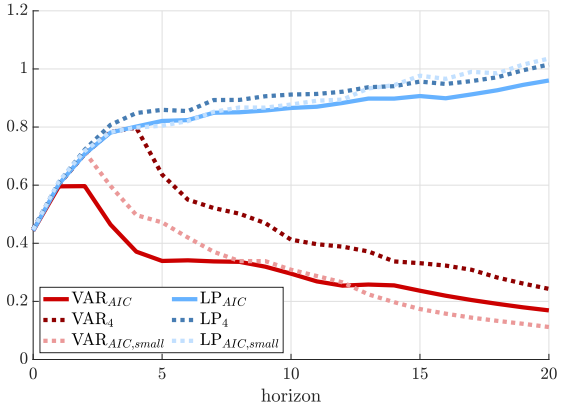
Standard deviation

average across 200 stationary and 200 non-stationary DGPs

Simulation evidence: bias and standard deviation



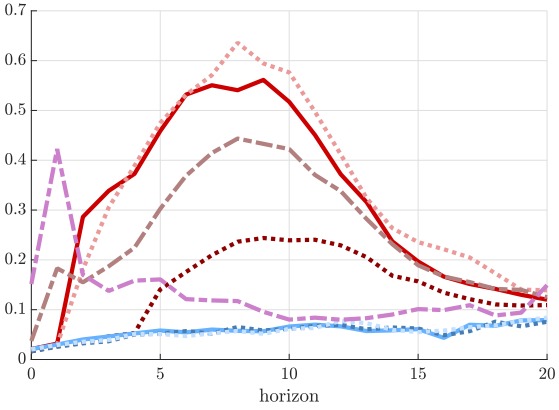
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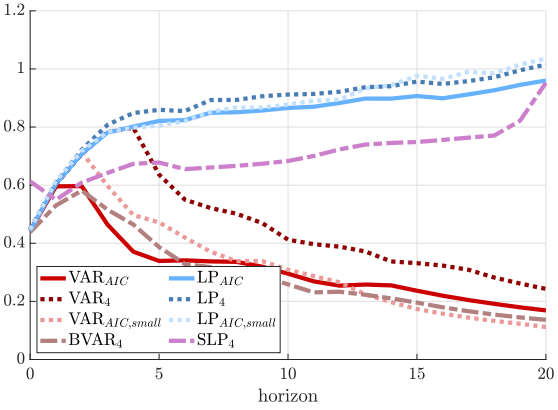
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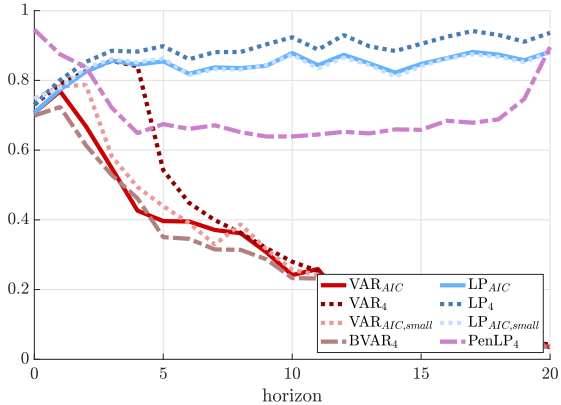


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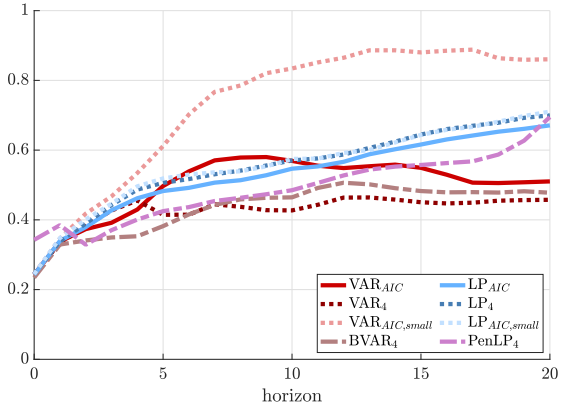
average across 200 stationary and 200 non-stationary DGPs

MSE loss: (B)VAR preferred over LP on average

Conventional way to trade off bias and variance: MSE = bias^2 + variance



MSE for stationary DGPs



MSE for non-stationary DGPs

Bias-variance trade-off: recap

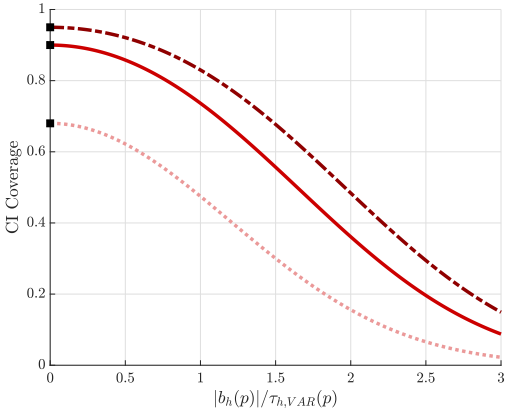
- Take-away: bias-variance trade-off is stark in practice.
- Robustness of LP to dynamic misspecification comes at significant variance cost.
- Under MSE loss, VAR is preferred over LP in the *average* simulation DGP.
  - Shrinkage (penalized LP or BVAR) often preferred over OLS.
- But MSE only evaluates the accuracy of the point estimate. This is not worth much without an accompanying uncertainty assessment.

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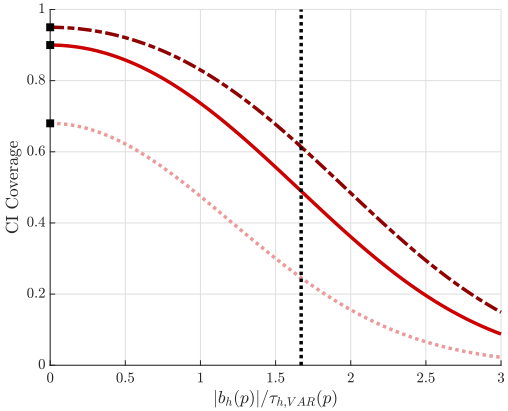
Uncertainty assessments: bias is costly

- Conventional to summarize uncertainty using confidence interval.
- Want coverage probability close to (say) 90% *regardless* of true DGP (not just for avg DGP!).
- Challenge for VARs: bias is really costly for coverage. CI has correct width, but off-center.
- Remember: easy to get worst-case bias $\approx 1.73 \times$ SE.

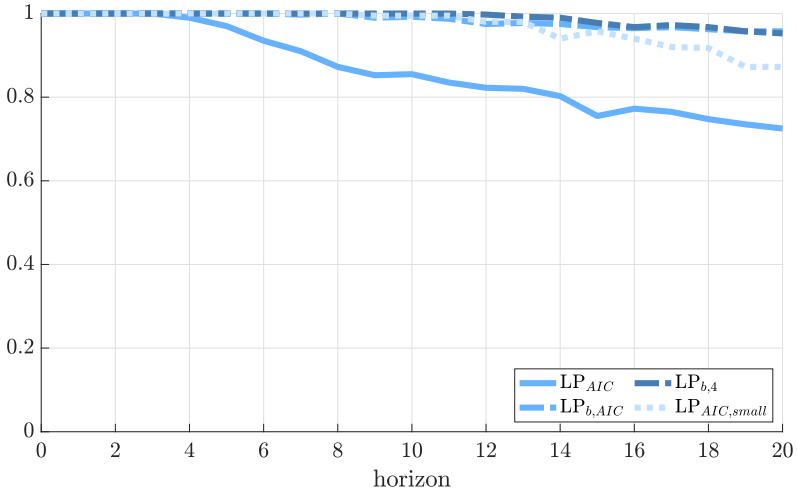


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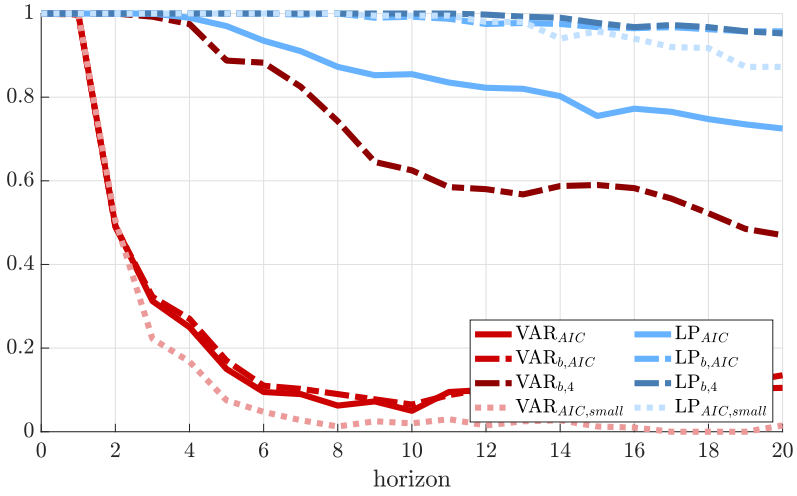


Simulation evidence: confidence interval coverage



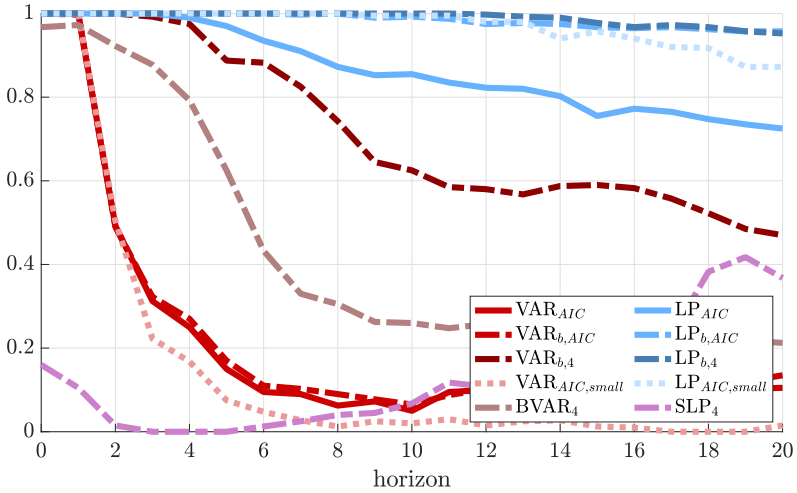
Fraction of DGPs with coverage $\geq$ 80% (target coverage 90%)

Simulation evidence: confidence interval coverage



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Simulation evidence: confidence interval coverage



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Long horizons and persistent data

- If the true DGP is a finite-order VAR, then the SVAR estimator is certainly efficient.
- But even in such settings, LP has some advantages: it's robust to persistence in the data and the length of the impulse response horizon h.
- Persistence:
  - The behavior of VAR estimators can depend sensitively on whether the data has unit roots (stochastic trends). Sometimes we get non-normally distributed estimators.
  - But the LP estimator projects on an (implicit) shock $\tilde{x}_t$, which is nicely stationary if we control for lags.
  - The LP estimator is therefore robustly (approximately) normally distributed, so the usual confidence interval works regardless of unit roots or not.

Long horizons and persistent data

- Long horizons:
  - The VAR impulse response formula is nonlinear in the VAR coefficients.
  - E.g., AR(1): $\hat{\theta}_h^{\text{VAR}} = \hat{\rho}^h$.
  - At horizons h that are a substantial fraction of the sample size T, the nonlinearity is so severe that standard VAR confidence intervals break down.
  - But the LP estimator is just based on linear regression. No nonlinearity issues, even for large h.

What did we discuss?

- Choice of VAR vs. LP has nothing to do with identification.
  - Anything you can do with SVAR, you can do with LP, and *vice versa*.
  - Proxy case: LP-IV is equivalent with "internal instrument" SVAR.
- Bias-variance trade-off between LP (low bias) and VAR (low variance).
  - No free lunch for VARs.
  - MSE loss: VAR (or BVAR) with few lags preferred for "typical" DGPs.
- Uncertainty assessments that are reliable across a wide range of DGPs and horizons require either:
  1. LP (controlling for lags, as we've discussed).
  2. VAR with many more lags than typically used in the literature. Use LP as robustness check.